Linear equations in two variables — Part 2

Algebra · Math · 65 questions

2.1 55d9f495-d3f8-4300-aadb-4beb4b5365d2 · Medium

What is the slope of the graph of $y = \frac{1}{4}27x + 15 + 7x$ in the xy-plane?

Student-produced response: _____

2.2 803e7a6a-e220-4562-862c-82f2d02fcb7f · Medium

Line t in the xy-plane has a slope of $-\frac{1}{3}$ and passes through the point $9, 10$. Which equation defines line t?

A. $y = 13x - \frac{1}{3}$

B. $y = 9x + 10$

C. $y = -\frac{x}{3} + 10$

D. $y = -\frac{x}{3} + 13$

2.3 4e5c1cee-ad79-4a5a-8828-24762395ae21 · Medium

A neighborhood consists of a 2-hectare park and a 35-hectare residential area. The total number of trees in the neighborhood is $3{,}934$. The equation $2x + 35y = 3{,}934$ represents this situation. Which of the following is the best interpretation of x in this context?

A. The average number of trees per hectare in the park

B. The average number of trees per hectare in the residential area

C. The total number of trees in the park

D. The total number of trees in the residential area

2.4 2764f11e-ac8f-4a9c-b6b0-db45ba9e4e48 · Medium

A total of 2 squares each have side length r. A total of 6 equilateral triangles each have side length t. None of these squares and triangles shares a side. The sum of the perimeters of all these squares and triangles is 210. Which equation represents this situation?

A. $6r + 24t = 210$

B. $2r + 6t = 210$

C. $8r + 18t = 210$

D. $6r + 2t = 210$

2.5 b0ba00cd-ad0e-4693-864f-2a0283cf64f7 · Medium

In the xy-plane, line s passes through the point $0, 0$ and is parallel to the line represented by the equation $y = 18x + 2$. If line s also passes through the point $4, d$, what is the value of d?

A. 2

B. 18

C. 72

D. 74

2.6 7840df29-5b0a-4ae4-9fb1-c8f458684f0b · Medium

Line k is defined by $y = 7x + \frac{1}{8}$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

A. -8

B. $-\frac{1}{7}$

C. $\frac{1}{8}$

D. 7

2.7 9ac48e4f-1444-4974-8049-572f31e5ed81 · Medium

If the graph of $27x + 33y = 297$ is shifted down 5 units in the *xy*-plane, what is the *y*-intercept of the resulting graph?

A. $0, 4$

B. $0, 6$

C. $0, 14$

D. $0, 28$

2.8 d85fc288-4098-4fba-a899-571c900d8231 · Medium

When line *n* is graphed in the *xy*-plane, it has an *x*-intercept of $-4, 0$ and a *y*-intercept of $0, \frac{86}{3}$. What is the slope of line *n*?

A. $\frac{3}{344}$

B. $\frac{6}{43}$

C. $\frac{43}{6}$

D. $\frac{344}{3}$

2.9 021851-DC · Medium

In an article about exercise, it is estimated that a 160-pound adult uses 200 calories for every 30 minutes of hiking and 150 calories for every 30 minutes of bicycling. An adult who weighs 160 pounds has completed 1 hour of bicycling. Based on the article, how many hours should the adult hike to use a total of 1,900 calories from bicycling and hiking?

A. 9.5

B. 8.75

C. 6

D. 4

2.10 ac7b345b-6b45-4f69-945c-5e7e16ca8909 · Medium

Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the xy-plane. What is the slope of line j?

A. $\frac{7}{17}$

B. $\frac{17}{7}$

C. 4

D. 17

2.11 3567f63b-262e-4925-aa9a-31ab8da38bd7 · Medium

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of $1{,}120$ points. The equation $80S + 90C = 1{,}120$ represents this situation. How many more points is a circle token worth than a square token?

A. 950

B. 90

C. 80

D. 10

2.12 efb2f017-fd60-4d55-aa5b-bf7aa344524d · Medium

x	y
-6	65
-3	56
3	38
6	29

The table shows four values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A. $9x + 3y = 141$

B. $9x + 3y = 3$

C. $3x + 9y = 141$

D. $3x + 9y = 3$

2.13 02a67ccb-75a5-4787-89d5-0516399af231 · Medium

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of $\$ 641$, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil, x, and the number of tons of crushed stone, y, Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

Student-produced response: _____

2.14 ab826e08-7891-4957-94db-0152ef740dff · Medium

Line p is defined by $2y + 18x = 9$. Line r is perpendicular to line p in the xy-plane. What is the slope of line r?

A. -9

B. $-\frac{1}{9}$

C. $\frac{1}{9}$

D. 9

2.15 b4b428b9-5d13-4880-8289-c67386f8fc40 · Medium

Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the xy-plane. What is the slope of line r?

A. -4

B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. 4

2.16 e7a189d8-ea3e-4bf1-b0c2-bb03d3f29337 · Medium

What is the slope of the graph of $10x - 5y = -12$ in the xy-plane?

A. -2

B. $-\frac{5}{6}$

C. $\frac{5}{6}$

D. 2

2.17 df2106f5-60f0-4169-a44d-b3223cb59788 · Medium

$$2x + y = 37$$

In triangle *QRS*, sides *QR* and *RS* each have a length of *x* centimeters and side *SQ* has a length of *y* centimeters. The given equation represents this situation. Which of the following is the best interpretation of 37 in this context?

A. The difference, in centimeters, between the lengths of sides *QR* and *SQ*

B. The difference, in centimeters, between the lengths of sides *QR* and *RS*

C. The sum of the lengths, in centimeters, of the three sides of the triangle

D. The length, in centimeters, of one of the two sides of equal length

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2.18 eb660cfe-6d9e-465b-a9a3-9b5a5ae60195 · Medium

$$2.5b + 5r = 80$$

The given equation describes the relationship between the number of birds, *b*, and the number of reptiles, *r*, that can be cared for at a pet care business on a given day. If the business cares for 16 reptiles on a given day, how many birds can it care for on this day?

A. 0

B. 5

C. 40

D. 80

2.19 6b671d8f-d733-464f-b0d9-9d825d5fd66c · Medium

The equation $7g + 7b = 840$ represents the number of blue tiles, *b*, and the number of green tiles, *g*, an artist needs for an 840-square-inch tile project. The artist needs 71 blue tiles for the project. How many green tiles does he need?

Student-produced response: _____

2.20 eb3d5dbd-4820-4685-bc1e-0710105586f6 · Medium

A store sells two different-sized containers of blueberries. The store's sales of these blueberries totaled 896.86 dollars last month. The equation $4.51x + 6.07y = 896.86$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. According to the equation, what is the price, in dollars, of each smaller container?

Student-produced response: _____

2.21 d739e9dd-c317-4c86-831a-07375a627a3d · Medium

A line in the xy-plane has a slope of 9 and passes through the point $0, -5$. The equation $y = px + r$ defines the line, where p and r are constants. What is the value of p?

Student-produced response: _____

2.22 10508e20-26b1-4e5b-8b38-3ab2f8695a5f · Medium

x	1	2	3
y	11	16	21

The table shows three values of x and their corresponding values of y. Which equation represents the linear relationship between x and y?

A. $y = 5x + 6$

B. $y = 5x + 11$

C. $y = 6x + 5$

D. $y = 6x + 11$

2.23 e4e2c39c-307d-411e-a829-b1a79fa0bee3 · Medium

A chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters mL. The volume of acetic acid in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

Student-produced response: _____

2.24 1ede31ff-90cd-4739-bd83-5ba88c2b02cb · Medium

Line *r* is defined by the equation $4x - 9y = 3$. Line *s* is parallel to line *r* in the *xy*-plane. What is the slope of line *s*?

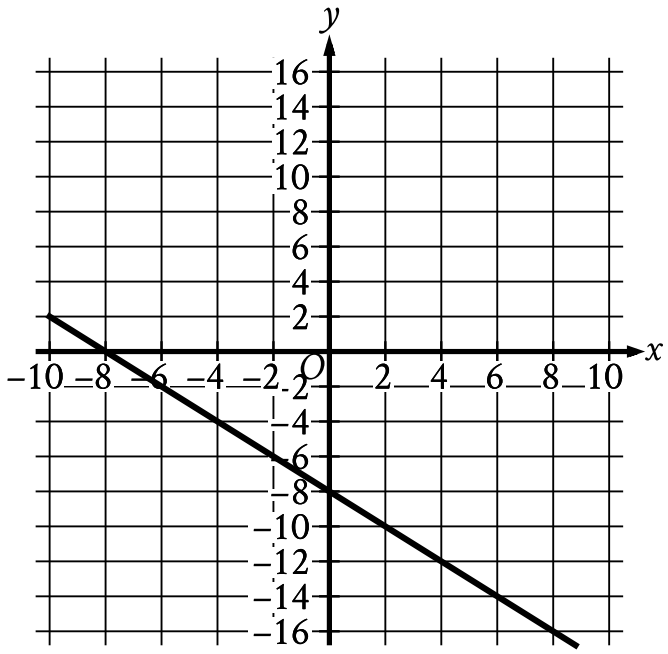
A. $\frac{9}{4}$

B. $\frac{4}{9}$

C. -4

D. -9

2.25 0d9d21e3-8ce9-45b7-b90e-60ee11a2b37c · Medium



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (negative 8 comma 0)
  - (0 comma negative 8)

What is an equation of the graph shown?

A. $y = -2x - 8$

B. $y = x - 8$

C. $y = -x - 8$

D. $y = 2x - 8$

2.26 f751d621-537a-4175-801e-a9a856b72c00 · Medium

What is the slope of the graph of $y = \frac{5x}{13} - 23$ in the *xy*-plane?

Student-produced response: _____

2.27 f33b0055-46c2-4d92-8dbc-03f057ed1886 · Medium

In the *xy*-plane, line *t* passes through the points $0, 9$ and $1, 17$. Which equation defines line *t*?

A. $y = \frac{1}{8}x + 9$

B. $y = x + \frac{1}{8}$

C. $y = x + 8$

D. $y = 8x + 9$

2.28 08073-DC · Medium

An artist paints and sells square tiles. The selling price P, in dollars, of a painted tile is a linear function of the side length of the tile s, in inches, as shown in the table below.

Side length, s (inches)	Price, P (dollars)
3	8.00
6	18.00
9	28.00

Which of the following could define the relationship between s and P ?

A. $P = 3s + 10$

B. $P = \frac{10}{3}s + 8$

C. $P = \frac{10}{3}s - 2$

D. $P = \frac{3}{10}s - \frac{1}{10}$

2.29 c9e742be-9c93-4d44-8ef4-0c15f7fff17f · Medium

Lily made 36 cups of jam. Lily then filled x small containers and y large containers with all the jam she made. The equation $4x + 6y = 36$ represents this situation. Which is the best interpretation of $6y$ in this context?

A. The number of large containers Lily filled

B. The number of small containers Lily filled

C. The total number of cups of jam in the large containers

D. The total number of cups of jam in the small containers

2.30 2f6fe99e-a285-4b30-b61f-c4864a95e6e6 · Medium

What is the slope of the graph of $y = \frac{1}{3}29x + 10 + 5x$ in the xy-plane?

Student-produced response: _____

2.31 02dc7dfd-b457-45d1-8655-2bcfcc4dfc17 · Medium

In the xy-plane, line k passes through the points $0, -5$ and $1, -1$. Which equation defines line k?

A. $y = -x + \frac{1}{4}$

B. $y = \frac{1}{4}x - 5$

C. $y = -x + 4$

D. $y = 4x - 5$

2.32 f7d44282-a880-4219-88e0-2d034cd8f06b · Medium

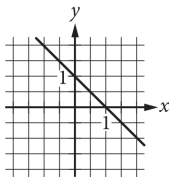
A line passes through the points $4, 6$ and $15, 24$ in the xy-plane. What is the slope of the line?

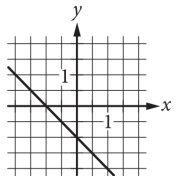
Student-produced response: _____

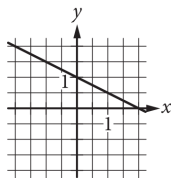
2.33 029477-DC · Hard

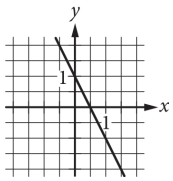
$ax + by = b$

In the equation above, a and b are constants and $0 < a < b$. Which of the following could represent the graph of the equation in the xy-plane?

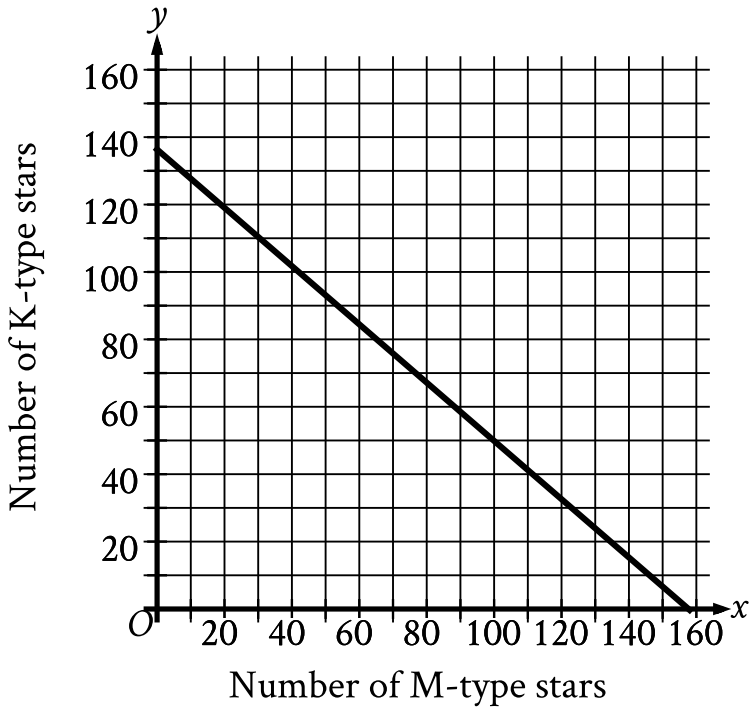
A. 

B. 

C. 

D. 

2.34 43b52d01-006d-4dbc-b7a8-3187b73ccbce · Hard



- The line slants gradually down from left to right.
- The line passes through the following approximate points:
  - (0 comma 136)
  - (158 comma 0)

A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is $127{,}882$ quettagrams. The graph shown models the possible combinations of the number of M-type stars, x, and K-type stars, y, that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

A. 811

B. 938

C. $51{,}904$

D. $75{,}978$

2.35 cdc3a36c-aceb-4d6e-99f2-44a62cd1b9b1 · Hard

$$5G + 45R = 380$$

At a school fair, students can win colored tokens that are worth a different number of points depending on the color. One student won G green tokens and R red tokens worth a total of 380 points. The given equation represents this situation. How many more points is a red token worth than a green token?

Student-produced response: _____

2.36 029556-DC · Hard

The graph of the equation $ax + ky = 6$ is a line in the xy-plane, where a and k are constants. If the line contains the points $(-2, -6)$ and $(0, -3)$, what is the value of k ?

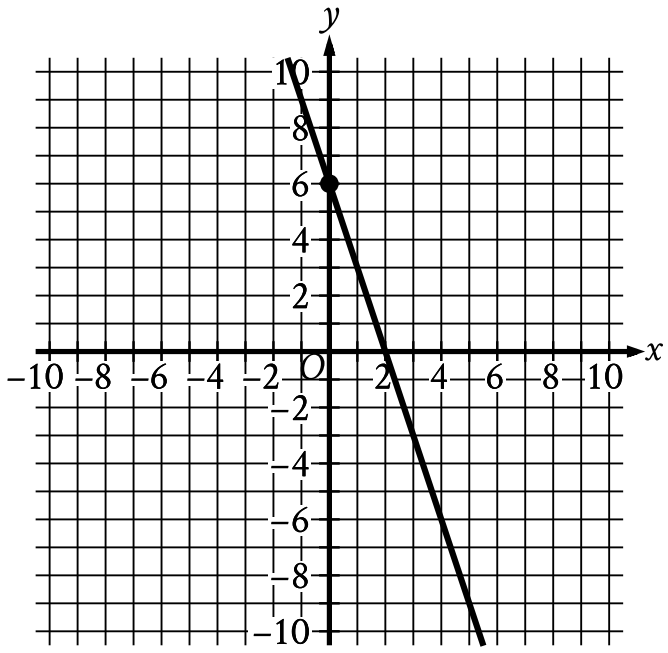
A. -2

B. -1

C. 2

D. 3

2.37 8cda2479-6500-45ad-b2fe-a4863fff3443 · Hard



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (0 comma 6)
  - (1 comma 3)

The graph shows a linear relationship between x and y. Which equation represents this relationship, where R is a positive constant?

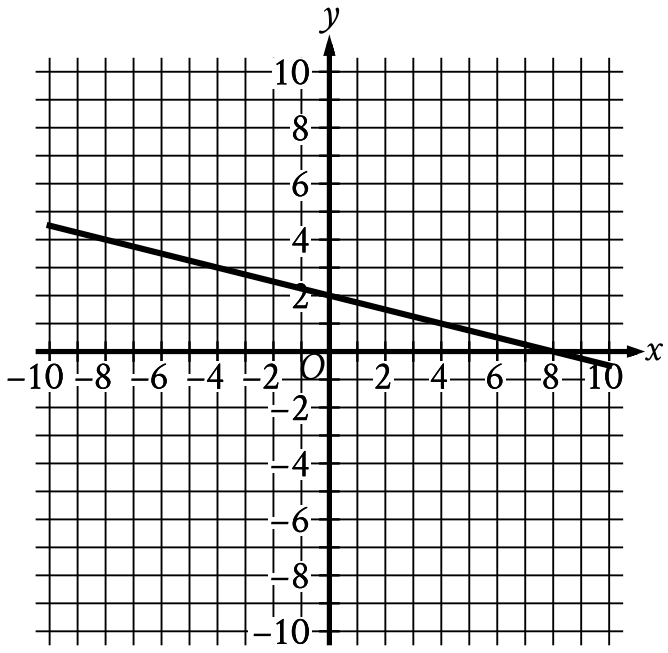
A. $Rx + 18y = 36$

B. $Rx - 18y = -36$

C. $18x + Ry = 36$

D. $18x - Ry = -36$

2.38 658ecb3f-b141-4d7a-abcb-105abef728e2 · Hard



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (negative 8 comma 4)
  - (0 comma 2)
  - (8 comma 0)

The graph of $y = fx + 14$ is shown. Which equation defines function f?

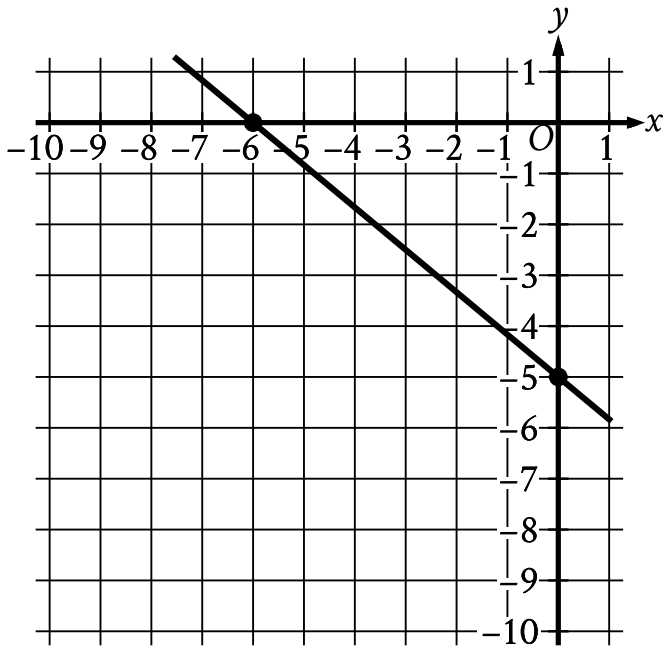
A. $fx = -\frac{1}{4}x - 12$

B. $fx = -\frac{1}{4}x + 16$

C. $fx = -\frac{1}{4}x + 2$

D. $fx = -\frac{1}{4}x - 14$

2.39 ee187260-bfb5-410e-bb49-e34aec865b7b · Hard



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (negative 6 comma 0)
  - (0 comma negative 5)

Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k. What is the slope of line j?

Student-produced response: _____

2.40 eb64b8f6-731a-4137-9ed2-358ab19f106c · Hard

x	y
-6	$n + 184$
-3	$n + 92$
0	n

The table shows three values of x and their corresponding values of y, where n is a constant, for the linear relationship between x and y. What is the slope of the line that represents this relationship in the xy-plane?

A. $-\frac{92}{3}$

B. $-\frac{3}{92}$

C. $\frac{n+92}{-3}$

D. $\frac{2n-92}{3}$

2.41 7e535b93-6192-4963-87af-547e39bbdfcf · Hard

x	y
18	130
23	160
26	178

For line h, the table shows three values of x and their corresponding values of y. Line k is the result of translating line h down 5 units in the xy-plane. What is the x-intercept of line k?

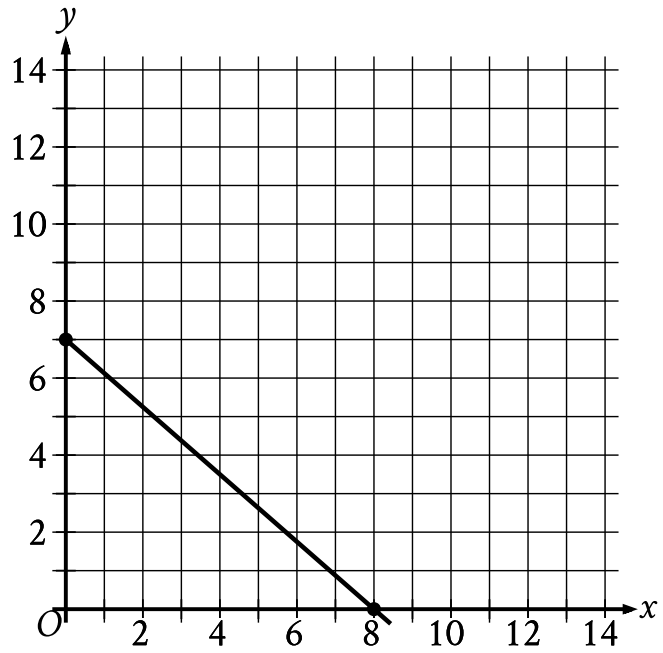
A. $-\frac{26}{3}, 0$

B. $-\frac{9}{2}, 0$

C. $-\frac{11}{3}, 0$

D. $-\frac{17}{6}, 0$

2.42 2af9fdfa-146c-4add-84b4-cb5bbf516dc2 · Hard



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (0 comma 7)
  - (8 comma 0)

The point with coordinates (d, 4) lies on the line shown. What is the value of d?

A. $\frac{7}{2}$

B. $\frac{26}{7}$

C. $\frac{24}{7}$

D. $\frac{27}{8}$

2.43 06cf762f-e23b-41cd-9a4f-cba279a432ff · Hard

What is the y-coordinate of the y-intercept of the graph of $\frac{3x}{7} = -\frac{5y}{9} + 21$ in the xy-plane?

Student-produced response: _____

2.44 c6aceeba-b507-44a2-bc31-9beb10096c93 · Hard

The graph of $7x + 2y = -31$ in the xy-plane has an x-intercept at $(a, 0)$ and a y-intercept at $(0, b)$, where a and b are constants. What is the value of $\frac{b}{a}$?

A. $-\frac{7}{2}$

B. $-\frac{2}{7}$

C. $\frac{2}{7}$

D. $\frac{7}{2}$

2.45 10934-DC · Hard

Line ℓ in the xy-plane is perpendicular to the line with equation $x = 2$. What is the slope of line ℓ?

A. 0

B. $-\frac{1}{2}$

C. -2

D. The slope of line ℓ is undefined.

2.46 0f40e0e9-fc67-443e-989b-f56c60365045 · Hard

$$0.10x + 0.20y = 0.17x + y$$

The equation gives a volume x, in gallons, of a 10% solution that could be mixed with a volume y, in gallons, of a 20% solution to produce a 17% solution. According to this equation, what volume, in gallons, of the 20% solution could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

Student-produced response: _____

2.47 80b79c80-6158-42f6-b5e5-0c0a19031fce · Hard

Line h is defined by $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$. Line j is perpendicular to line h in the xy-plane. What is the slope of line j?

A. $-\frac{7}{5}$

B. $-\frac{5}{7}$

C. $\frac{7}{5}$

D. $\frac{5}{7}$

2.48 029626-DC · Hard

x	y
3	7
k	11
12	n

The table above shows the coordinates of three points on a line in the xy-plane, where k and n are constants. If the slope of the line is 2, what is the value of $k+n$?

Student-produced response: _____

2.49 d4043654-05ea-4067-80ff-0c1687134c44 · Hard

A certain apprentice has enrolled in 85 hours of training courses. The equation $10x + 15y = 85$ represents this situation, where x is the number of on-site training courses and y is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

Student-produced response: _____

2.50 2b8c0dff-e373-422a-a150-3f281e461330 · Hard

Line l is defined by $3y + 12x = 5$. Line n is perpendicular to line l in the xy-plane. What is the slope of line n?

Student-produced response: _____

2.51 0741234a-eaa5-4593-8dae-f0666e86a222 · Hard

x	y
$-2s$	24
$-s$	21
s	15

The table shows three values of x and their corresponding values of y, where s is a constant. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A. $sx + 3y = 18s$

B. $3x + sy = 18s$

C. $3x + sy = 18$

D. $sx + 3y = 18$

2.52 83c4d031-e8f1-4415-8c6c-33d11d65e85b · Hard

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy-plane. The y-intercept of the line is $k - 5, b$, where k and b are constants. What is the value of b?

Student-produced response: _____

2.53 b7ba7dde-8eb4-42fb-a134-796966b15f0d · Hard

x	y
-18	-48
7	52

The table shows two values of x and their corresponding values of y. In the xy-plane, the graph of the linear equation representing this relationship passes through the point $\frac{1}{7}, \quad a$. What is the value of a?

A. $-\frac{4}{11}$

B. $-\frac{4}{77}$

C. $\frac{4}{7}$

D. $\frac{172}{7}$

2.54 070614-DC · Hard

How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?

Student-produced response: _____

2.55 045840-DC · Hard

In the xy-plane, line k is defined by $x+y=0$. Line j is perpendicular to line k, and the y-intercept of line j is $(0,3)$. Which of the following is an equation of line j ?

A. $x+y=3$

B. $x+y=-3$

C. $x-y=3$

D. $x-y=-3$

2.56 8f4b667c-7537-4495-ba4d-4549d1e62f85 · Hard

$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy-plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A. $10x + 7y = 1$ $ax - 2by = 1$

B. $10x + 7y = 1$ $ax + 2by = 1$

C. $10x + 7y = 1$ $2ax + by = 1$

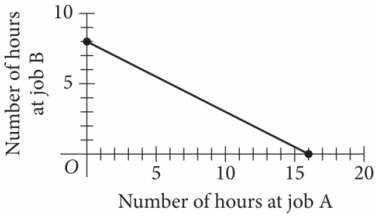
D. $5x - 7y = 1$ $ax + by = 1$

2.57 155657c1-cb62-489d-b3be-d87bbbf5aada · Hard

In the xy-plane, line l passes through the point $0, 0$ and is parallel to the line represented by the equation $y = 8x + 2$. If line l also passes through the point $3, d$, what is the value of d?

Student-produced response: _____

2.58 029469-DC · Hard



To earn money for college, Avery works two part-time jobs: A and B. She earns $10 per hour working at job A and $20 per hour working at job B. In one week, Avery earned a total of s dollars for working at the two part-time jobs. The graph above represents all possible combinations of numbers of hours Avery could have worked at the two jobs to earn s dollars. What is the value of s ?

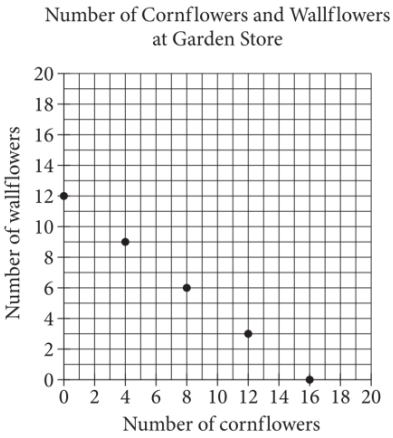
A. 128

B. 160

C. 200

D. 320

2.59 070612-DC · Hard



The points plotted in the coordinate plane above represent the possible numbers of wallflowers and cornflowers that someone can buy at the Garden Store in order to spend exactly $24.00 total on the two types of flowers. The price of each wallflower is the same and the price of each cornflower is the same. What is the price, in dollars, of 1 cornflower?

Student-produced response: _____

2.60 070613-DC · Hard

The line with the equation $\frac{4}{5}x + \frac{1}{3}y = 1$ is graphed in the xy-plane. What is the x-coordinate of the x-intercept of the line?

Student-produced response: _____

2.61 02e0a325-790e-4a20-956c-c646b0e6a227 · Hard

The graph of $9x - 10y = 19$ is translated down 4 units in the *xy*-plane. What is the *x*-coordinate of the *x*-intercept of the resulting graph?

Student-produced response: _____

2.62 a44626e2-3239-48f1-90f6-145e3b8b2c54 · Hard

Keenan made 32 cups of vegetable broth. Keenan then filled *x* small jars and *y* large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

A. The number of large jars Keenan filled

B. The number of small jars Keenan filled

C. The total number of cups of vegetable broth in the large jars

D. The total number of cups of vegetable broth in the small jars

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2.63 3e65e30b-6f1f-4e34-9ac7-2f2019988ed3 · Hard

$$\frac{3}{5}x + \frac{3}{4}y = 7$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
1	$\frac{113}{20}$
2	$\frac{101}{20}$
4	$\frac{77}{20}$

B.

x	y
1	$\frac{47}{5}$
2	$\frac{44}{5}$
4	$\frac{38}{5}$

C.

x	y
1	$\frac{148}{15}$
2	$\frac{136}{15}$
4	$\frac{112}{15}$

D.

x	y
1	$\frac{128}{15}$
2	$\frac{116}{15}$

4	$\frac{92}{15}$

2.64 6e2c4ed8-a7d5-4552-bdd7-42223af298ea · Hard

Line *p* is defined by $4y + 8x = 6$. Line *r* is perpendicular to line *p* in the *xy*-plane. What is the slope of line *r*?

Student-produced response: _____

2.65 029630-DC · Hard

In the xy-plane, line k intersects the y-axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line k, what is the value of w ?

Student-produced response: _____